Nabanita Sarker

nabanitataylor@gmail.com | 01568120879 | Nabanita Sarker | NabanitaSarker162

Education

BSc **Bangladesh University of Engineering and Technology** — January 2022 (ongoing)

Electrical and Electronic Engineering

Major: Electronics and Photonics

- CGPA: 3.70/4.00 (up to Level 4, Term 1)
- **Relevant Coursework:** VLSI Circuits and Design, Optoelectronics, Semiconductor Processing & Fabrication, Microprocessor and Embedded Systems, Analog Integrated Circuits and Designs, Solid State Devices, Compound Semiconductor Devices, Introduction to nanotechnology and nanoelectronics

HSc **Holy Cross College** — July 2018- June 2020

Science

- GPA: 5.00/5.00

Research Interests

2D materials   VLSI Design   Nanoelectronics   Neuromorphic Computing

Research Experience

Enhanced photocatalytic hydrogen and oxygen evolution activity by two-dimensional vanderWaals heterostructure: A first-principles study — April 2025 (ongoing)

- Investigating the electronic, optical, and photocatalytic properties of van der Waals (vDW) heterostructure using density functional theory
- Studying heterostructure as an excellent water splitting photocatalyst with remarkable potential for fuel cells and energy storage applications

Professional Experience

Udvash, Chemistry Instructor — Dhaka, Bangladesh (ongoing)

Academic Projects

High-Performance and Low-Power CMOS Implementation of a Password-Protected Car Parking System Using Cadence Design Flow — Report

- The system supports secure slot selection, password storage and verification, parking time measurement, charge calculation, and alarm generation for incorrect password attempts. Special emphasis is placed on performance optimization (high speed, low latency), power minimization, and maximum area utilization during PnR.
- Tools Used: Cadence

Development of a Color Intensity Based Approach for pH Estimation Using Optical Sensing — Report

- This project is about the design and implementation of a low-cost optical colorimetric pH sensing system based on LED–photodiode technology. The system determines pH by analyzing the change in color intensity of a pH indicator solution using optical absorption principles. An Arduino-based data acquisition and processing unit is used to convert optical signals into calibrated pH values, which are displayed on a computer interface.

- Tools Used: Arduino, MATLAB

Common-Base RF Amplifier for High-Frequency Applications Report

- Designed an RF Amplifier with required specifications of Frequency: 5 GHz, Gain greater than 8 dB PAE greater than 0.05, Psat greater than 10 dBm
- Tools Used: Cadence

IoT Based Real Time Intravenous Fluid Monitoring and Flow Control System Report

- Developed a monitoring system that continuously tracks fluid level in the IV bag, Automatically regulates flow rate or allows manual control by doctors/nurses via IoT interface, Sends alerts/notifications to a connected application in case of Low fluid level, Patient abnormalities (e.g., fever irregular pulse rate), Irregular flow rate
- Tools Used: Blynk, Autodesk Fusion360, Arduino

8 Player Quiz Game Machine Report

- Built a Machine that consists of a central quiz unit, multiple input devices (buttons) for each player, and a visual output display for presenting questions, answer choices, and player scores without microcontroller
- Tools Used: Proteus

Versatile CNC Machine: Pen-plotter, Cutter, Engraver Report

- Designed a CNC machine that not only draws but also can also be used in cutting and engraving purpose
- Tools Used: CNC shield and arduino

Battery Charge Controller using Buck Converter with Auto Cut-off Feature Report

- The project is about designing a battery charger with variable DC output using a Buck converter and an auto cutoff feature
- Tools Used: Proteus

Real Time Fingerprint Recognition using Matlab Report

- The goal of this project is to develop a complete system for real time data collection and fingerprint verification through extracting and matching minutiae
- Tools Used: MATLAB

Technical Skills

Hardware: Microcontrollers, IoT Devices, Sensors.

DFT Tools: Biovia Materials Studio, Quantum Espresso, VESTA.

Circuit Simulation and Design: Cadence, Pspice, LTSpice, Proteus.

Others: C, C++, AutoCad, MATLAB, AutoDesk Fusion 360, LaTex

Honors and Awards

University Merit Scholarship

General Scholarship in Secondary School Certificate (SSC) Examination

Talentpool Scholarship in Junior School Certificate (JSC) Examination

Perfect Attendance Certificate, Holy Cross College, Recognized for flawless attendance.

References

Dr. Shaikh Asif Mahmood

- Professor, Department of Electrical and Electronic Engineering
- Bangladesh University of Engineering and Technology
- Email: asifmahmood@eee.buet.ac.bd

Dr. Mahbub Alam

- Professor, Department of Electrical and Electronic Engineering
- Bangladesh University of Engineering and Technology
- Email: mahbubalam@eee.buet.ac.bd